Module 1 Challenge

Calculus Honors · The Idea of a Limit · 16 points

Name: _____ Date: _____

> This is your fullest, most careful work of the module. Show every step in clean vertical layout, justify your reasoning in words where asked, and label your final answers. You will be graded on understanding, quality of reasoning, clarity and precision, and intellectual honesty (say clearly where you are certain and where you are not).

1. Build and explain (5 pts). Consider $f(x) = \dfrac{x^2 - x - 6}{x - 3}$. (a) Build a two-sided table with three values approaching 3 from each side and estimate $\lim_{x \to 3} f(x)$. (b) Explain in two or three sentences what your table shows and why the limit exists even though $f(3)$ is undefined.

2. Resolve with algebra (6 pts). Evaluate $\lim_{x \to 0} \dfrac{\sqrt{x + 9} - 3}{x}$ exactly. Show the rationalizing step in full, and state which limit laws justify your final substitution.

3. One-sided reasoning (5 pts). Let $g(x) = \begin{cases} x + 1 & x < 2 \\ 5 - x & x \geq 2 \end{cases}$. Find $\lim_{x \to 2^-} g(x)$, $\lim_{x \to 2^+} g(x)$, and $g(2)$. Decide whether $\lim_{x \to 2} g(x)$ exists, and justify your decision. Note clearly how confident you are in each answer.